NATIONAL INSTITUTE OF TECHNOLOGY TIRUCHIRAPPALLI

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Eye-Gaze Typing on a Commodity Webcam

A Cluster-Constrained Hybrid Decoder for Accessible Text Entry

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Why Type with the Eyes?

Typing assumes working hands
Motor impairments, sterile or hands-busy settings — the keyboard quietly excludes many users.

Precise gaze needs costly hardware
Reliable eye-trackers use infrared rigs that rarely leave the lab or clinic.

A webcam sees gaze only coarsely
From a plain camera, the gaze estimate can be off by an entire key width.

“The eyes already point at what we intend to select. Recover that point well enough, and looking becomes typing.”

THE OBJECTIVE

A complete eye-typing system on a standard webcam — no added hardware — that never outputs text the user did not gaze.

From a Glance to a Keystroke

1. **Webcam frame**
   one ordinary video frame
2. **Face & iris landmarks**
   where each iris sits in the eye
3. **8 geometric features**
   iris ratios + head position
4. **Calibrated mapping**
   personal fit → screen point
5. **Dwell to commit**
   hold 0.69 s → key types

Per-user calibration

Look at 7 screen spots and click each — that fits a small personal gaze model. Every later click quietly refines it, so accuracy improves with use.

The Midas-touch problem

Eyes always point somewhere — every glance would type. Dwell turns looking into a deliberate act: a key fires only after a held fixation, then briefly re-arms.

The Keyboard — Direct Gaze Typing



Full-screen keys — one exact letter each

Typing **h e l** |

help | hello | held

↑ dwell a suggestion to finish the word

Dwell to type

Rest your gaze on a key; hold about 0.69 s and that exact letter is entered — a ring fills as live feedback.

Exact by design

One letter per key, so what you look at is what you type — no decoding step, and no wrong words.

Word autocomplete

A prefix pulls the top words from a 16,839-word vocabulary; dwell one to finish the whole word in a single gaze.

The research frontier: fewer, larger keys are easier to gaze — but each holds several letters, turning a word into a short, ambiguous code. Resolving that ambiguity, faithfully, is the decoder's job.

The Cluster-Constrained Hybrid Decoder

1 · Tier 1 · Exact lookup

An unambiguous code maps to its single known word — deterministic, correct by construction.

2 · Tier 2 · Constrained lattice search

Ambiguous codes expand to candidate words; a compact language model picks the most fluent path. Every path re-encodes to the input.

3 · Tier 3 · Neural disambiguation

A sequence model proposes full sentences — accepted only if they re-encode exactly; otherwise it merely re-ranks the lattice paths.

If a code was never seen and no consistent sentence exists, the result is flagged — never silently guessed.

THE INVARIANT

Every accepted output re-encodes to exactly the code that was gazed.

Why it matters

A fluent-but-wrong sentence cannot slip through: the model disambiguates, but can never hallucinate.

What Was Built — and Verified

Delivered, end to end

- In-browser gaze tracking, calibration and dwell keyboard
- The constrained decoder over a 16,839-word vocabulary built from ~173k sentences
- A reproducible corpus → vocabulary → model → evaluation pipeline

Verified behaviour

- Encoding round-trips and every decoder tier — 23 checks, all passing
- Full typing flow driven end-to-end in the browser — 16 checks
- Dwell threshold confirmed behaviourally, feedback synchronised

478
landmarks per frame

8
gaze features

0.69 s
dwell to commit

16,839
word vocabulary

Honest scope: real-gaze and corpus-level accuracy are not yet measured — the evaluation harness is ready, and that measurement is the immediate next step.

Takeaways & What's Next

Typing without hands, hardware or guesswork

A standard webcam and a browser are enough for gaze-driven text entry.

A decoder that cannot contradict the user

Language models disambiguate under a hard consistency constraint — no hallucinated words.

Reproducible by design

One shared encoding across data, training and decoding; an evaluation harness ready to run.

NEXT

- Quantitative accuracy evaluation
- Adaptive dwell time to reduce fatigue
- A study with the users it is built for

Thank you · Questions welcome

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